UNIVERSITÀ DEGLI STUDI DI PADOVA
Department of Civil, Environmental and Architectural Engineering

Master's Degree in Mathematical Engineering
Mathematical Modelling for Engineering and Science

Master's Thesis

Investigations on some models in gradient elastoplasticity

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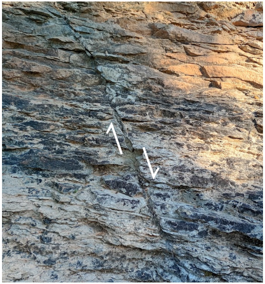
Academic Year 2025/2026

Introduction

Illustration of necking from A. Considère's mémoire *"Mémoire sur l'emploi du fer et de l'acier dans les constructions"* (1885).



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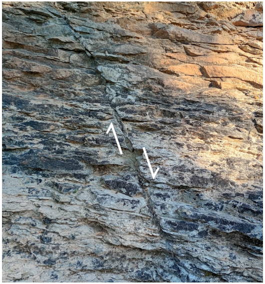

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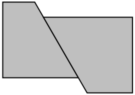
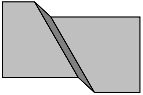


Non-gradient elastoplasticity



Gradient elastoplasticity

Some introductory notions

Developing mechanical models through the principle of virtual powers

Description of a mechanical system

The systems are the elements z of a separable and reflexive Banach space Z.

Generalized forces are the elements $\mathcal{F}$ of the dual space Z^*. Given a virtual rate $w \in Z$ the corresponding power is $\langle \mathcal{F}, w \rangle$.

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Equilibrium

Equilibrium is the balance

$$-\mathcal{I}(t) + \mathcal{X}(t) = 0 \quad \text{for a.e. } t \in (0, T)$$

between internal forces $\mathcal{I} \colon (0, T) \to Z^*$ and external forces $\mathcal{X} \colon (0, T) \to Z^*$.

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Constitutive relations

Let $\Psi\colon Z \to \mathbb{R} \cup \{0\}$ be the free energy and $\Phi\colon Z \to [0, \infty]$ the dissipation potential.

Internal forces are related to the evolution $z\colon [0, T] \to Z$ with a constitutive relation

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$$\mathcal{I}(t) \in D\Psi(z(t)) + \partial\Phi(\dot{z}(t)) \quad \text{for a.e. } t \in (0,T).$$

Definition. $\xi \in \partial\Phi(v)$ if $\Phi(w) \geqslant \Phi(v) + \langle \xi, w - v \rangle$ for all $w \in Z$.

Proposition. If Φ is convex and $\Phi(0) = 0$ then $\langle \xi, v \rangle \geqslant 0$ for all $\xi \in \partial\Phi(v)$.

Corollary. $\frac{d}{dt}\Psi(z(t)) \leqslant \langle \mathcal{I}(t), \dot{z}(t) \rangle = \langle \mathcal{X}(t), \dot{z}(t) \rangle$.

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Problem. Given $z_0 \in Z$ and $\mathcal{X}$ find z such that $z(0) = z_0$ and equilibrium holds.

$\Longrightarrow$

Doubly nonlinear inclusion

$$D\Psi(z(t)) + \partial\Phi(\dot{z}(t)) \ni \mathcal{X}(t)$$

for a.e. $t \in (0, T)$

Modeling dilatant and pressure sensitive elastoplastic materials

Classical approach

Kinematics

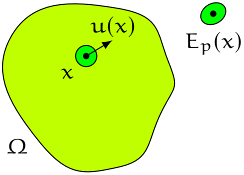
The space of configurations is $Z = \{z = (u, E_p)\}$ with

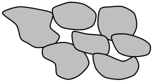
the displacement $u \in H^1(\Omega, \mathbb{R}^d)$

the plastic strain $E_p \in H^1(\Omega, \mathrm{Sym})$.

The elastic strain is $E_e = Eu - E_p$.

The generic rate is $w = (v, V_p)$ and the elastic rate is $V_e = Ev - V_p$.





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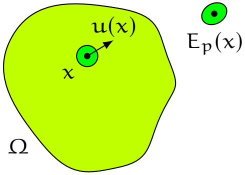
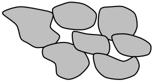




Statics

The internal virtual power is

$$\langle \mathcal{I}, w \rangle = \int_\Omega \left[T_e : V_e + T_p : V_p \right] dx$$

with $T_e \colon \Omega \to \mathrm{Sym}$ the elastic stress, $T_p \colon \Omega \to \mathrm{Sym}$ the plastic force. The external virtual power is

$$\langle \mathcal{X}, w \rangle = \int_\Omega f \cdot v \, dx + \int_{\partial\Omega} h \cdot v \, dx.$$

Proposition. The strong form of the equilibrium is

$$\left.\begin{aligned} \operatorname{div} T_e + f &= 0 \\ T_e - T_p &= 0 \end{aligned}\right. \quad \text{in } \Omega$$

and $T_e n = h$ in $\partial\Omega$.

Classical approach

Constitutive relations

Let the free energy

$$\Psi(z) = \int_\Omega \frac{1}{2}\mathfrak{C}E_e : E_e\,dx,$$
$$\text{so that} \quad T_e = \mathfrak{C}E_e.$$

Yield condition and non-associated flow rule

Fix $f, g\colon \mathrm{Sym} \to \mathbb{R}$ convex. Let

$$C = \{A \in \mathrm{Sym} | f(A) \leqslant 0\}$$
$$B(T_p) = \{A \in \mathrm{Sym} | g(A) \leqslant g(T_p)\}.$$

Impose that $T_p \in C$. The yield condition is $T_p \in \partial C$ and the flow rule is

$$\dot{E}_p \in \begin{cases} \{0\} & \text{if } T_p \in \operatorname{int} C \\ N_{B(T_p)}(T_p) & \text{if } T_p \in \partial C. \end{cases}$$

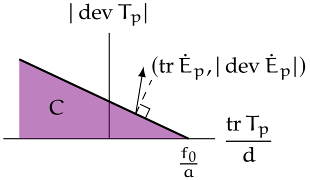
Proposition. If $f = g$, is equivalent to $T_p \in \partial\sigma_C(\dot{E}_p)$ with $\sigma_C(\dot{E}_p) = \sup_{T_p \in C} T_p : \dot{E}_p$ so that

$$\Phi(w) = \int_\Omega \sigma_C(V_p)\,dx.$$

Example. Drucker–Prager yield criterion

$$f(T_p) = |\operatorname{dev} T_p| - \left(f_0 - a\frac{\operatorname{tr} T_p}{d}\right)$$
$$g(T_p) = |\operatorname{dev} T_p| + b\frac{\operatorname{tr} T_p}{d}$$

with $0 \leqslant b \leqslant a$.



The flow rule $\dot{E}_p = \gamma \frac{\partial g}{\partial T_p}(T_p)$ implies

$$\operatorname{dev} \dot{E}_p = \gamma \frac{\operatorname{dev} T_p}{|\operatorname{dev} T_p|}, \quad \operatorname{tr} \dot{E}_p = b|\operatorname{dev} \dot{E}_p|$$

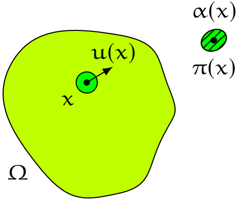
with $\gamma f(T_p) = 0$, $f(T_p) \leqslant 0$ and $\gamma \geqslant 0$.

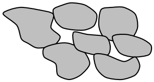
Kinematics

The space of configurations is $Z = \{z = (u, \pi, \alpha, \lambda)\}$

- the displacement $u \in H^1(\Omega, \mathbb{R}^d)$
- the deviatoric plastic strain $\pi \in H^1(\Omega, \mathrm{DevSym})$
- the damage variable $\alpha \in H^1(\Omega)$
- the volumetric plastic strain $\lambda \in L^2(\Omega)$

Let $E_p = \pi + \frac{\lambda}{d} I$ and $E_e = Eu - E_p$.





Proposal of an alternative approach

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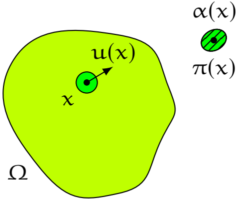
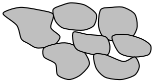
Notation for virtual rates

The generic rate is $w = (v, \rho, \beta, \mu)$, the plastic rate is $V_p = \rho + \frac{\mu}{d} I$ and the elastic rate is $V_e = Ev - V_p$.





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Addition of a constrain

Add the constrain $z \in K = \{\lambda = L(\alpha)\}$ with $L \colon \mathbb{R} \to \mathbb{R}$

$$\int_\Omega (\lambda - L(\alpha)) q \, dx = 0 \quad \text{for all } q \in L^2(\Omega).$$

Micropressure

Parametrize the constraint reaction $\mathcal{P} \in -N_K(z)$ as

$$\langle \mathcal{P}, w \rangle = \int_\Omega p(\mu - L'(\alpha)\beta) \, dx$$

with $p \in L^2(\Omega)$ the micropressure.

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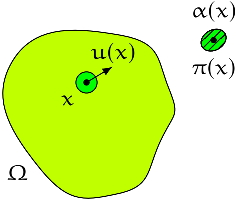
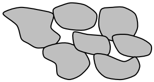




EXAMPLE. $L(\alpha) = b\,\alpha_0 \left[1 - \exp\left(-\frac{\alpha}{\alpha_0}\right)\right]$

Proposal of an alternative approach

Statics

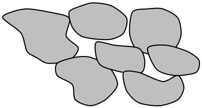
The internal virtual power is

$$\langle \mathcal{I}, w\rangle = \int_\Omega \left[T : E v + K \vdots \operatorname{grad}^2 v \right] dx + \\ + \int_\Omega \left[T_e : V_e + T_p : \rho + K_p \vdots \operatorname{grad} \rho \right] dx + \\ + \int_\Omega \left[\tau \beta + \kappa \cdot \operatorname{grad} \beta \right] dx$$

with $T \colon \Omega \to \mathrm{Sym}$ the macroscopic stress, $K \colon \Omega \to \mathrm{Lin} \otimes \mathbb{R}$ the macroscopic hyperstress, $T_e \colon \Omega \to \mathrm{Sym}$ the elastic stress, $T_p \colon \Omega \to \mathrm{DevSym}$ the plastic force, $K_p \colon \Omega \to \mathrm{DevSym} \otimes \mathbb{R}$ the plastic stress, $\tau \colon \Omega \to \mathbb{R}$ the damage force and $\kappa \colon \Omega \to \mathbb{R}^d$ the damage stress. The external virtual power is

$$\langle \mathcal{X}, w\rangle = \int_\Omega f \cdot v \, dx + \int_{\partial\Omega} h \cdot v \, dx.$$

The equilibrium is $-\mathcal{I} + \mathcal{X} + \mathcal{P} = 0$.



Proposition. The strong form of the equilibrium is

$$\begin{gathered} \operatorname{div}(T + T_e - \operatorname{div} K) + f = 0 \\ \operatorname{dev} T_e - T_p + \operatorname{div} K_p = 0 \\ -\tau + \operatorname{div} \kappa - L'(\alpha) p = 0 \\ p = -\frac{\operatorname{tr} T_e}{d} \end{gathered}$$

in Ω and $T_e n = h$, $K_p n = 0$, $\kappa \cdot n$ in $\partial\Omega$.

Proposal of an alternative approach

Constitutive relations

Let the free energy be $\Psi(z) = \frac{1}{2}\langle \mathcal{A}z, z\rangle + \Psi_0(z)$ with

$$\frac{1}{2}\langle \mathcal{A}z, z\rangle = \frac{1}{2}\int_\Omega \Big[\mathfrak{a}|\operatorname{grad}^2 u|^2 + \\ + \mathfrak{b}|\operatorname{grad}\pi|^2 + \\ + \mathfrak{c}|\operatorname{grad}\alpha|^2\Big]\,dx$$

and

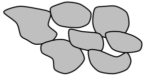
$$\Psi_0(z) = \int_\Omega \left[\frac{1}{2}\mathfrak{C}E_e : E_e + W_d(\alpha)\right] dx.$$

Let the dissipation potential be $\Phi(p, w) = \frac{1}{2}\langle \mathcal{D}h, h\rangle + \Phi_0(p, h)$ with

$$\frac{1}{2}\langle \mathcal{D}h, h\rangle = \frac{1}{2}\int_\Omega \left[\mathfrak{D}Ev : Ev + \mathfrak{d}|\rho|^2 + \mathfrak{e}|\beta|^2\right] dx$$

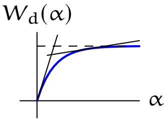
and

$$\Phi_0(p, h) = \int_\Omega [R(p, \rho) + Q(\rho, \beta)]\,dx.$$



Example.

$$W_d(\alpha) = f_0\alpha_1\left[1 - \exp\left(-\frac{\alpha}{\alpha_1}\right)\right]$$



$$R(p, \rho) = Yp^+|\rho|$$

$$Q(\rho, \beta) = \begin{cases} 0 & \text{if } \beta \geqslant |\rho| \\ \infty & \text{if } \beta < |\rho|. \end{cases}$$

Remark. This impose $\dot\alpha \geqslant |\dot\pi|$ so that $\dot\lambda \geqslant L'(\alpha)\dot\alpha|\dot\pi|$.

Recovering the classical model, well-posedness result and numerics

Well-posedness result

Existence, uniqueness and continuous dependence on the data for the constrained doubly nonlinear equation

$$
\begin{aligned}
&D_y \Psi_0(z(t)) + \mathcal{A}y(t) + \\
&+ \partial_h \Phi_0(p(t), \dot{y}(t)) + \mathcal{D}\dot{y}(t) - \\
&\qquad - \mathcal{L}'(y(t))^* p(t) \ni \mathcal{X}(t)
\end{aligned}
$$

$$D_\lambda \Psi_0(z(t)) + p(t) = 0$$

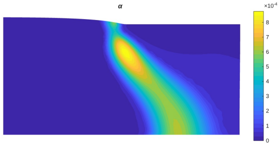
$$\lambda(t) = \mathcal{L}(y(t)).$$

Recovering the classical approach

If suitable constitutive relations are chosen then

our approach $\Longrightarrow$ Drucker–Prager model

Numerics

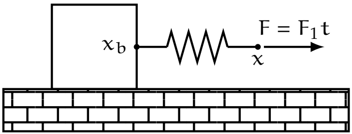


Thank you for your attention.

Further explanations of the general theoretical framework

A simple example

A block on the plane with friction pulled with a spring.



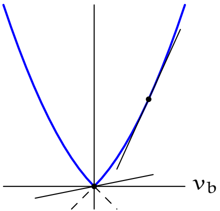
The space of configurations is $Z = \{z = (x, x_b) \in \mathbb{R}^2\}$. Let $w = (v, v_b)$ the generic rate of the system.

The free energy is $\Psi(z) = \frac{\kappa}{2}|x - x_b|^2$ while the dissipation potential is $\Phi(w) = Y|v_b| + \frac{\eta}{2}|v_b|^2$.

Start with $z(0) = (0, 0)$.

Proposition. It holds that

$$\partial\Phi(v_b) = \begin{cases} [-Y, Y] & \text{if } v_b = 0 \\ \{Y \operatorname{sign} v_b + \eta v_b\} & \text{if } v_b \neq 0. \end{cases}$$



Solution. It holds that $x = x_b + \frac{F}{\kappa}$. Moreover $\dot{x}_b = 0$ when $F < Y$ while $F = Y + \eta\dot{x}_b$ when $F \geqslant Y$.

Treating constraints with the method of Lagrange multipliers

Constrained kinematics

Let $K \subset Z$ closed and convex and impose the constraint $z(t) \in K$ for all $t \in [0, T]$.

Reaction force

Add a reaction force $\mathcal{P}\colon (0, T) \to Z^*$ satisfying $\mathcal{P}(t) \in -N_K(z(t)) = -\partial\iota_K(z(t))$ for a.e. $t \in (0, T)$.

Equilibrium

The equilibrium now reads

$$-\mathcal{J}(t) + \mathcal{X}(t) + \mathcal{P}(t) = 0 \quad \text{for a.e. } t \in (0, T).$$

Constitutive relations

Let again $\Psi\colon Z \to \mathbb{R} \cup \{0\}$ be the free energy.

Now the dissipation potential $\Phi\colon Z^* \times Z \to [0, \infty]$, $(\mathcal{P}, w) \mapsto \Phi(\mathcal{P}, w)$ can depend on the reaction force.

The constitutive relations become

$$\mathcal{J}(t) \in D\Psi(z(t)) + \partial_w \Phi(\mathcal{P}(t), \dot{z}(t))$$

for a.e. $t \in (0, T)$.

Problem. Given $z_0 \in Z$ and $\mathcal{X}$ find z and $\mathcal{P}$ such that $z(0) = z_0$, $z(t) \in K$ for all $t \in [0, T]$, $\mathcal{P}(t) \in -N_K(z(t))$ for a.e. $t \in (0, T)$ and equilibrium holds.

$\Longrightarrow$

Constrained doubly nonlinear inclusion

$$\begin{gathered} z(t) \in K \quad \text{for all } t \in [0, T] \\ \mathcal{P}(t) \in -N_K(z(t)) \\ D\Psi(z(t)) + \partial\Phi(\mathcal{P}(t), \dot{z}(t)) \ni \mathcal{X}(t) + \mathcal{P}(t) \end{gathered}$$

for a.e. $t \in (0, T)$

Further details on the models for dilatant and pressure sensitive elastoplastic materials

Recovering the classical Ducker–Prager model

PROPOSITION. Letting $L(\alpha) = b\,\alpha$, $W_d(\alpha) = f_0\alpha$, $R(p,\rho) = Yp|\rho|$ with $Y = a - b$ we recover the Drucker–Prager model.

SKETCH OF THE PROOF. For simplicity disregard viscous and gradient terms. Recall the balance

$$\begin{aligned} \operatorname{dev} T_e - T_p &= 0 \\ -\tau - L'(\alpha)p &= 0. \end{aligned}$$

This becomes

$$\begin{aligned} \operatorname{dev} T_e &\in (a-b)\partial|\cdot|(\dot\pi) + \partial_\rho Q(\dot\pi,\dot\alpha) \\ f_0 + bp &\in -\partial_\beta Q(\dot\pi,\dot\alpha). \end{aligned}$$

The choice $Y = a - b$ weights correctly the contributions of plastic deviatoric and damage resistance to provide the right yield condition

$$|\operatorname{dev} T_e| \leqslant f_0 + ap.$$

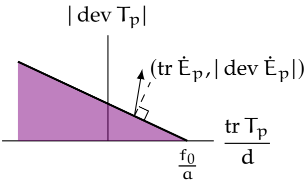
The flow rule is satisfied since $\dot\lambda = b\,\dot\alpha = b|\dot\pi|$. □

RECALL. Drucker–Prager yield criterion

$$f(T_p) = |\operatorname{dev} T_p| - \left(f_0 - a\frac{\operatorname{tr} T_p}{d}\right)$$

$$g(T_p) = |\operatorname{dev} T_p| + b\frac{\operatorname{tr} T_p}{d}$$

with $0 \leqslant b \leqslant a$.



The flow rule $\dot E_p = \gamma\frac{\partial g}{\partial T_p}(T_p)$ implies

$$\operatorname{dev} \dot E_p = \gamma\frac{\operatorname{dev} T_p}{|\operatorname{dev} T_p|}, \quad \operatorname{tr} \dot E_p = b|\operatorname{dev} \dot E_p|$$

with $\gamma f(T_p) = 0$, $f(T_p) \leqslant 0$ and $\gamma \geqslant 0$.

Recovering the classical Ducker–Prager model

Proposition. Letting $L(\alpha) = b\,\alpha$, $W_d(\alpha) = f_0\,\alpha$, $R(p,\rho) = Yp|\rho|$ with $Y = a - b$ we recover the Drucker–Prager model.

Sketch of the proof. For simplicity disregard viscous and gradient terms. Recall the balance

$$\begin{aligned} \operatorname{dev} T_e - T_p &= 0 \\ -\tau - L'(\alpha)p &= 0. \end{aligned}$$

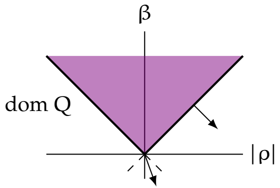
This becomes

$$\begin{aligned} \operatorname{dev} T_e &\in (a-b)\partial|\cdot|(\dot\pi) + \partial_\rho Q(\dot\pi,\dot\alpha) \\ f_0 + bp &\in -\partial_\beta Q(\dot\pi,\dot\alpha). \end{aligned}$$

The choice $Y = a - b$ weights correctly the contributions of plastic deviatoric and damage resistance to provide the right yield condition

$$|\operatorname{dev} T_e| \leqslant f_0 + ap.$$

The flow rule is satisfied since $\dot\lambda = b\,\dot\alpha = b|\dot\pi|$. □



Lemma. When $\dot\pi \neq 0$ and $\dot\alpha = |\dot\pi|$ the generic element of $\partial Q(\dot\pi,\dot\alpha)$ is $\left(-\theta\frac{\dot\pi}{|\dot\pi|}, \theta\right)$ with $\theta \leqslant 0$. When $\dot\pi = 0$ and $\dot\alpha = 0$ is (Θ,θ) with $\theta \leqslant 0$ and $\Theta \in \mathrm{DevSym}$ with $|\Theta| \leqslant |\theta|$.

Notation

Define

$$Y = \{y = (u, \pi, \alpha)\}, \ \Lambda = \{\lambda\}, \ \Lambda^* = \{p\}.$$

and $\mathcal{L}\colon Y \to \Lambda$ by $\mathcal{L}(y) = L(\alpha)$. Denote the virtual rate as $w = (h, \mu)$ with $h = (v, \rho, \beta) \in Y$ and $\mu \in \Lambda$. Then

$$\lambda = \mathcal{L}(y)$$
$$\langle \mathcal{P}, w \rangle = \langle p, \mu \rangle - \langle \mathcal{L}'(y)^* p, h \rangle.$$

Existence and uniqueness theorem

Let $H = \{y = (u, \pi, \alpha)\}$ with $u \in H^1(\Omega, \mathbb{R}^d)$, $\pi \in L^2(\Omega, \mathrm{DevSym})$ and $\alpha \in L^2(\Omega)$.

Assume that

- L is bounded, differentiable and with bounded and Lipschitz derivative
- W_d is differentiable with Lipschitz derivative
- $R(p, \rho) = Y(p)\tilde{R}(\rho)$ with Y positive and Lipschitz, $\tilde{R}\colon \mathrm{DevSym} \to [0, \infty)$ convex, Lipschitz and $\tilde{R}(0) = 0$
- $Q\colon \mathrm{DevSym} \times \mathbb{R} \to [0, \infty]$ is positive convex, l.s.c. Lipschitz on its domain and $Q(0, 0) = 0$.

Theorem. There exist a unique solution $z = (y, \lambda)$ and p with $z(0) = 0$, $y \in W^{1,\infty}((0, T); Y)$, $\dot{y} \in C^0([0, T], H)$, $\lambda \in C^0([0, T], \Lambda)$ and $p \in C^0([0, T], \Lambda^*)$. Moreover these quantities depend continuously on the data with respect to the norms of their spaces.

Formulation of the problem

Assume homogeneous boundary conditions. Given $f \in H^1((0,T), H^{-1}(\Omega, \mathbb{R}^d))$ let

$$\langle \mathcal{X}(t), h \rangle = \langle f(t), v \rangle$$

PROBLEM. Find $z = (y, \lambda)\colon [0,T] \to Z$ and $p\colon [0,T] \to \Lambda^*$ such that $z(0) = 0$ and

$$D_y \Psi_0(z(t)) + \mathcal{A} y(t) + \\ + \partial_h \Phi_0(p(t), \dot{y}(t)) + \mathcal{D}\dot{y}(t) - \\ - \mathcal{L}'(y(t))^* p(t) \ni \mathcal{X}(t)$$

$$D_\lambda \Psi_0(z(t)) + p(t) = 0$$

$$\lambda(t) = \mathcal{L}(y(t))$$

for a.e. $t \in (0,T)$.

Existence and uniqueness theorem

Let $H = \{y = (u, \pi, \alpha)\}$ with $u \in H^1(\Omega, \mathbb{R}^d)$, $\pi \in L^2(\Omega, \mathrm{DevSym})$ and $\alpha \in L^2(\Omega)$.

Assume that

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THEOREM. There exist a unique solution $z = (y, \lambda)$ and p with $z(0) = 0$, $y \in W^{1,\infty}((0,T); Y)$, $\dot{y} \in C^0([0,T], H)$, $\lambda \in C^0([0,T], \Lambda)$ and $p \in C^0([0,T], \Lambda^*)$. Moreover these quantities depend continuously on the data with respect to the norms of their spaces.

Abstract existence and uniqueness for constrained doubly nonlinear equations

Assumptions

Let $Z = Y \times \Lambda$ with Y, Λ separable and reflexive Banach spaces. Let H a separable and reflexive and Banach space such that $Y \subset H$ with compact and dense inclusion and

$$c_P\|y\|_Y \leqslant \|y\|_H + |y|_Y \leqslant C_P\|y\|_Y.$$

Let $\mathcal{L}\colon H \to \Lambda$ bounded, differentiable with bounded and Lipschitz differential.

Let $\Psi(z) = \Psi_0(z) + \frac{1}{2}\langle \mathcal{A}y, y\rangle$ and $\Phi(p,h) = \Phi_0(p,h) + \frac{1}{2}\langle \mathcal{D}h, h\rangle$ with

- $\Psi_0\colon H \times \Lambda \to \mathbb{R}$ Frèchet differentiable with Lipschitz differential
- $\mathcal{A} \in \mathcal{L}(Y, Y^*)$ self-adjoint and $\langle \mathcal{A}y, y\rangle \geqslant C_A|y|_V^2$
- $\Phi_0\colon \Lambda^* \times H \to [0,\infty]$ with $\Phi_0(p,\cdot)$ convex, $\Phi_0(p,0) = 0$, $|\Phi_0(p,h_1) - \Phi_0(p,h_2)| \leqslant C_\Phi(\|p\|_\Lambda + 1)\|h_1 - h_2\|_H$, $|\Phi_0(p_1,h) - \Phi_0(p_2,h)| \leqslant C_\Phi\|p_1 - p_2\|_{\Lambda^*}\|h\|_H$, $|\Phi_0(p_1,h_1) - \Phi_0(p_1,h_2) + \Phi_0(p_2,h_2) - \Phi_p(p_2,h_1)| \leqslant C_\Phi\|p_1 - p_2\|_{\Lambda^*}\|h_1 - h_2\|_H$
- $\mathcal{D} \in \mathcal{L}(H, H^*)$ self-adjoint and $\langle \mathcal{D}h, h\rangle \geqslant C_D\|h\|_H^2$

Abstract existence and uniqueness for constrained doubly nonlinear equations

Formulation of the problem

Let $\mathcal{X} \in H^1((0,T), H^*)$ assigned.

Problem. Find $z = (y, \lambda)\colon [0,T] \to Z$ and $p\colon [0,T] \to \Lambda^*$ such that $z(0) = 0$ and

$$\begin{aligned} &D_y \Psi_0(z(t)) + \mathcal{A} y(t) + \\ &+ \partial_h \Phi_0(p(t), \dot{y}(t)) + \mathcal{D}\dot{y}(t) - \\ &\qquad - \mathcal{L}'(y(t))^* p(t) \ni \mathcal{X}(t) \end{aligned}$$

$$D_\lambda \Psi_0(z(t)) + p(t) = 0$$

$$\lambda(t) = \mathcal{L}(y(t))$$

for a.e. $t \in (0,T)$.

Existence and uniqueness theoem

Theorem. There exist a unique solution $z = (y, \lambda)$ and p with $z(0) = 0$, $y \in W^{1,\infty}((0,T); Y)$, $\dot{y} \in C^0([0,T], H)$, $\lambda \in C^0([0,T], \Lambda)$ and $p \in C^0([0,T], \Lambda^*)$. Moreover these quantities depend continuously on the data with respect to the norms of their spaces.

Sketch of the proof. Discretize in time.

$$\begin{aligned} &D_y \Psi_0(z_N^k) + \mathcal{A} y_N^k + \\ &+ \partial_h \Phi_0\left(p_N^k, \frac{y_N^k - y_N^{k-1}}{\tau_N}\right) + \mathcal{D}\frac{y_N^k - y_N^{k-1}}{\tau_N} - \\ &\qquad - \mathcal{L}'(y_N^k)^* p_N^k \ni \mathcal{X}_N^k \end{aligned}$$

$$D_\lambda \Psi_0(z_N^k) + p_N^k = 0$$

$$\lambda_N^k = \mathcal{L}(y_N^k).$$

Substitute the constraint in the first equation and consider the equation in y_N^k for each p_N^k fixed

$$\begin{aligned} &D_y \Psi_0(\mathcal{L}(y_N^k), y_N^k) + \mathcal{A} y_N^k + \\ &+ \partial_h \Phi_0\left(p_N^k, \frac{y_N^k - y_N^{k-1}}{\tau_N}\right) + \mathcal{D}\frac{y_N^k - y_N^{k-1}}{\tau_N} - \\ &\qquad - \mathcal{L}'(y_N^k)^* p_N^k \ni \mathcal{X}_N^k. \end{aligned}$$

Abstract existence and uniqueness for constrained doubly nonlinear equations

Is of the form $F(y) + \partial\Xi(y) \ni 0$ with $F\colon H \to H^*$ continuous and $\Xi\colon Y \to [0,\infty]$ convex. Coercivity

$$\lim_{\|y\|_Y \to \infty} \frac{\Xi(y) + \langle F(y), y\rangle}{\|y\|_Y} = \infty$$

if τ is small thanks to viscosity. Minty–Browder theorem ensures that the solution $y_N^k = \mathcal{U}(p_N^k)$ exists unique for all p_N^k.

The second equation becomes

$$p_N^k = -D_\lambda \Psi_0(\mathcal{L}(\mathcal{U}(p_N^k)), \mathcal{U}(p_N^k)) = \mathcal{T}(p_N^k).$$

Existence is established with Schauder fixed-point theorem.

The discrete solution has been constructed. Let $\bar{y}_N$, y_N be the piecewise constant and affine interpolations of y_N^k. Let $\hat{y}_N$ be the piecewise constant interpolation of $\dot{y}_N^k$.

A priori estimates gives y_N bounded in $W^{1,\infty}((0,T),Y)$ and $\hat{y}_N$ bounded in $L^2((0,T),H)$. By compactness

$$\begin{aligned} y_N &\overset{*}{\rightharpoonup} y \quad \text{in } W^{1,\infty}((0,T),Y)\\ y_N &\to y \quad \text{in } C^0([0,T],Y)\\ \bar{y}_N &\to y \quad \text{in } L^2((0,T),Y)\\ \hat{y}_N &\to \dot{y} \quad \text{in } L^2([0,T],H). \end{aligned}$$

Letting $\lambda_N = \mathcal{L}(y_N)$, $\bar{\lambda}_N = \mathcal{L}(\bar{y}_N)$, $\lambda = \mathcal{L}(y)$, $p_N = -D_\lambda\Psi_0(z_N)$, $\bar{p}_N = -D_\lambda\Psi_0(\bar{z}_N)$, $p = -D_\lambda\Psi_0(z)$,

$$\begin{aligned} \lambda_N &\to \lambda \quad \text{in } C^0([0,T],\Lambda)\\ \bar{\lambda}_N &\to \lambda \quad \text{in } L^2((0,T),\Lambda)\\ p_N &\to p \quad \text{in } C^0([0,T],\Lambda^*)\\ \bar{p}_N &\to p \quad \text{in } L^2((0,T),\Lambda^*). \end{aligned}$$

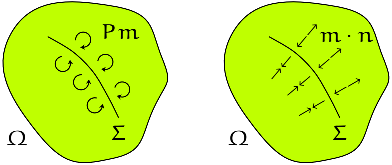
This allows to pass to the limit. □

Contact torque as a Lagrange multiplier

Let $\Sigma \subset \Omega$ be an oriented smooth surface. Remove the constraint of continuity of $\frac{\partial u}{\partial n}$ across Σ and impose it with the method of Lagrange multipliers

$$\int_\Sigma \left[\!\!\left[\frac{\partial u}{\partial n} \right]\!\!\right] \cdot l \, dx = 0, \quad \langle \mathcal{M}, h \rangle = \int_\Sigma m \cdot \left[\!\!\left[\frac{\partial v}{\partial n} \right]\!\!\right] dx.$$

for all $l \in H^{-1/2}(\Sigma, \mathbb{R}^d)$. Here $m \in H^{-1/2}(\Sigma, \mathbb{R}^d)$ is the contact torque.



Let $Y_\Sigma = \{y = (u, \pi, \alpha)\}$ with $u \in H^1(\Omega, \mathbb{R}^d) \cap H^2(\Omega \setminus \Sigma, \mathbb{R}^d)$ and π, α as before.

Problem. Find $z = (y, \lambda) \colon [0, T] \to Y_\Sigma \times \Lambda$ and $p \colon [0, T] \to \Lambda^*$ such that $z(0) = 0$ and

$$\begin{aligned} &D_y \Psi_0(z(t)) + \mathcal{A} y(t) + \\ &\quad + \partial_h \Phi_0(p(t), \dot{y}(t)) + \mathcal{D} \dot{y}(t) - \\ &\quad - \mathcal{L}'(y(t))^* p(t) \ni \mathcal{X}(t) + \mathcal{M}(t) \end{aligned}$$

$$D_\lambda \Psi_0(z(t)) + p(t) = 0$$

$$\lambda(t) = \mathcal{L}(y(t))$$

$$\left[\!\!\left[\frac{\partial u}{\partial n}(t) \right]\!\!\right] = 0 \quad \text{on } \Sigma$$

for a.e. $t \in (0, T)$.

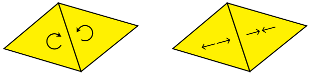
Numerical approximation

Finite element spaces

Fix a triangulation $\mathcal{T}_l = \{K\}$ and define $\Sigma_l = \Omega \cap \bigcup_{K \in \mathcal{T}_l} \partial K$. Let

$$\begin{aligned} u \in H^1_0(\Omega, \mathbb{R}^d) \cap H^2(\Omega \setminus \Sigma_l, \mathbb{R}^d) &\quad \text{P2} \\ \pi \in H^1(\Omega, \mathrm{DevSym})\} &\quad \text{P1} \\ \alpha \in H^1(\Omega) &\quad \text{P1} \\ \lambda \in L^2(\Omega) &\quad \text{P0} \\ p \in L^2(\Omega) &\quad \text{P0} \\ m \in H^{-1/2}(\Sigma_l, \mathbb{R}^d) &\quad \text{P0} \end{aligned}$$

and consider the finite element spaces $Y_l = \{y = (u, \pi, \alpha)\}$, $\Lambda_l = \{\lambda\}$, $P_l = \{p\}$ and $M_l = \{m\}$.



Numerical scheme

Fix time steps τ^k with $k \in \{1, \dots, N\}$.

Problem. Find $z^k = (y^k, \lambda^k) \in Y_l \times \Lambda_l$, $p^k_\tau \in P_l$ and $m^k \in M_l$ such that $z^0 = 0$ and

$$\Big\langle D_y \Psi_0(z^k) + \mathcal{A} y^k + \\ + D_h \Phi_0 \left(p^k, \frac{y^k - y^{k-1}}{\tau^k} \right) + \mathcal{D} \frac{y^k - y^{k-1}}{\tau^k} - \\ - \mathcal{L}'(y^k)^* p^k - \mathcal{X}^k - \mathcal{M}^k, h \Big\rangle = 0$$

$$\langle D_\lambda \Psi_0(z^k) + p^k, \mu \rangle = 0$$

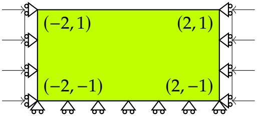
$$\langle \lambda^k - \mathcal{L}(y^k), q \rangle = 0$$

$$\left\langle l, \left[\!\left[\frac{\partial u^k}{\partial n} \right]\!\right] \right\rangle = 0$$

for all $h \in Y_l$, $\mu \in \Lambda_l$, $q \in P_l$, $l \in M_l$ and $k \in \{1, \dots, N\}$.

Validation

Validate comparing with an analytical solution.



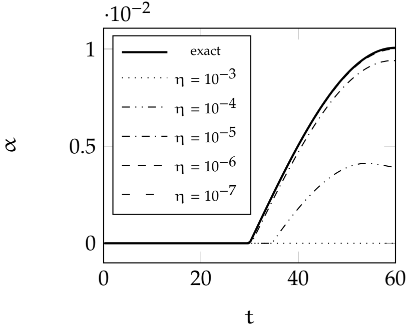
The solution is spatially homogeneous.

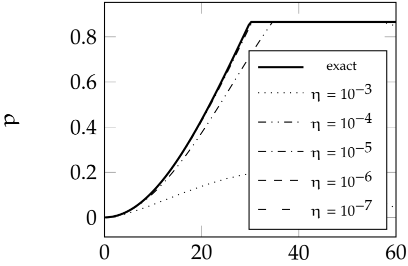
Regularize the dissipation potentials with the Yosida approximation

$$\tilde{R}_\eta(\rho) = \inf_{\xi \in \mathrm{DevSym}} \left[\frac{|\rho - \xi|^2}{2\eta} + |\xi| \right]$$

$$Q_\eta(\rho, \beta) = \frac{1}{2\eta} \mathrm{dist}_{\mathrm{dom}\, Q}(\rho, \beta)^2.$$

with $\eta > 0$.

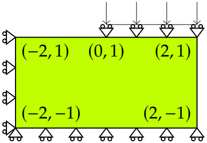


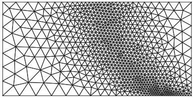


Numerical approximation

An example

The aim is to show the formation of a shear band.





Let $\zeta > 0$ a multiplier of the gradient coefficients $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$.

Choose

$$\mathfrak{a} = \sqrt{2} \sin \phi \quad \mathfrak{b} = \sqrt{2} \sin \psi \quad \mathfrak{f}_0 = \sqrt{2} \mathfrak{c} \cos \phi$$

with $\mathfrak{c} > 0$ the cohesion, $\phi, \psi \in [0, 90°)$ the friction and dilatancy angle.